

Chapter 6 Fluid resistance and energy loss

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Section 2 Two regimes/states of fluid motion

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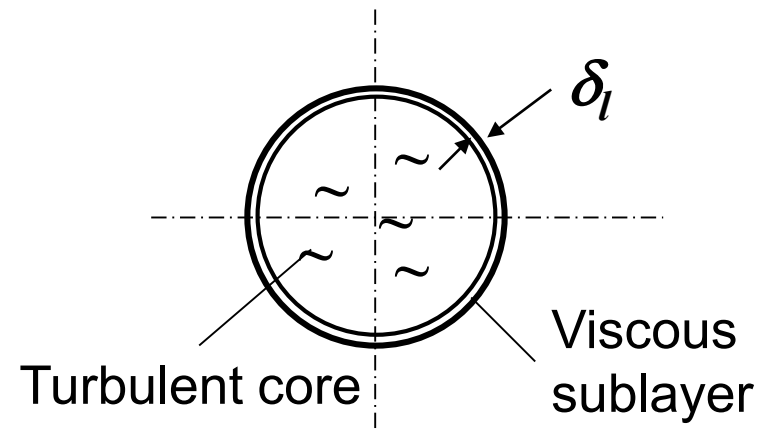
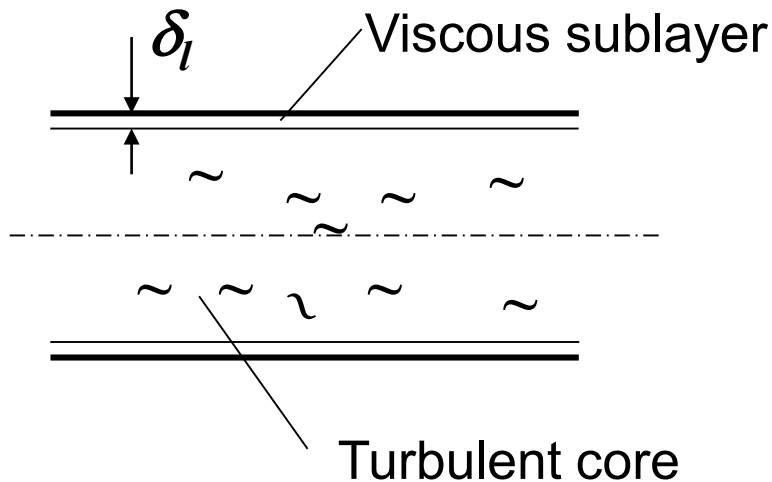
- ➡ 1. Structure of turbulence
- 2. Time average of turbulent motion
- 3. Reynolds stress in turbulent flow
- 4. Prandtl mixing length
- 5. Velocity distribution of turbulent flow

1. Structure of turbulence: viscous sublayer and turbulent core region

• Basic concepts

Viscous sublayer (粘性底层) : large velocity gradient, viscous effect non-negligible.

Turbulent core region (湍流核心区) : The fluid flow outside the viscous sublayer is called the turbulent core.



- Viscous sublayer

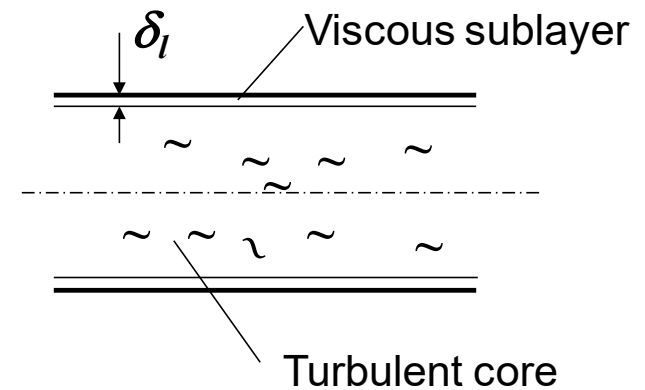
- a. Velocity distribution in viscous sublayer

Based on the Newton 's law of internal friction:

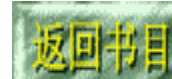
$$\tau_0 = \mu \frac{d\bar{u}_x}{dy} \approx \mu \frac{\bar{u}}{y}$$

$$u_*^2 = \frac{\tau_0}{\rho} = \frac{\mu}{\rho} \frac{\bar{u}}{y} = \nu \frac{\bar{u}}{y}$$

Therefore $\frac{\bar{u}}{u_*} = \frac{u_* y}{\nu} \quad (y \leq \delta_l)$



Conclusion: the velocity in the viscous sublayer is **linearly distributed** along y direction.



b. Thickness of viscous sublayer

Experimental data show that,

$$\text{when } y \rightarrow \delta_l, \quad \frac{u_* \delta_l}{\nu} = 11.6$$

$$\delta_l = 11.6 \frac{\nu}{u_*} = 11.6 \frac{\nu}{\sqrt{gRJ}} = \frac{32.8d}{Re\sqrt{\lambda}}$$

Re ——Flow Reynolds number in tube;

λ ——Frictional resistance coefficient

$$\frac{\bar{u}}{u_*} = \frac{u_* y}{\nu}$$

$$u_* = u \sqrt{\frac{\lambda}{8}} = \sqrt{gRJ}$$

Note:

(1) The viscous bottom is very thin, generally with only a few tenths of millimeter.

(2) With the pipe diameter d fixed, the Reynolds number increases and the viscous sublayer becomes thinner with the increase of flow velocity.



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2. Time average of turbulent motion

- Pulsation of turbulent motion quantities and their time average
- Time averaged velocity:

$$\bar{u}_x = \frac{1}{T} \int_0^T u_x(t) dt$$

$$u_x = \bar{u}_x + u'_x$$

$$\text{or } u_y = \bar{u}_y + u'_y, \quad u_z = \bar{u}_z + u'_z$$

$$p = \bar{p} + p'$$

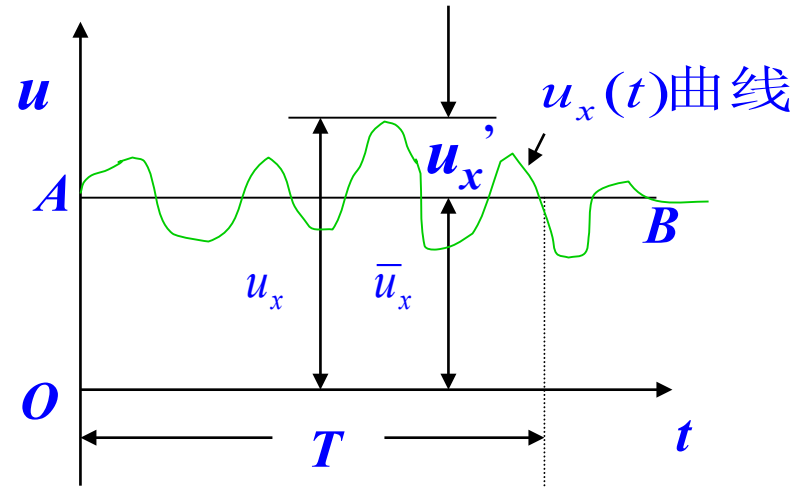
- Turbulent intensity

$$N = \frac{\sqrt{\frac{1}{3}(\overline{u_x'^2} + \overline{u_y'^2} + \overline{u_z'^2})}}{\bar{u}}$$

- Properties of fluctuation quantities:

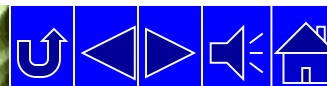
$$\overline{u'_x} = \frac{1}{T} \int_0^T u'_x dt = \frac{1}{T} \int_0^T (u_x - \bar{u}_x) dt = \bar{u}_x - \bar{u}_x = 0$$

$$\overline{u_x'^2} \neq 0, \quad \overline{u'_i u'_j} \neq 0$$



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3. Reynolds stress in turbulent flow

(1) Expression of turbulent shear stress

$$\tau = \tau_v + \tau_t = \mu \frac{d\bar{u}}{dy} + (-\rho \overline{u'_x u'_y})$$

Notes:

- 1) With small **Reynolds number**, turbulent pulsation is weak, and the **viscous shear stress** is dominant;
- 2) With large **Reynolds number**, turbulent pulsation and hence the turbulent shear stress is intensified. In the fully-developed turbulent core, the viscous shear stress is negligible compared to the **turbulent shear stress**;
- 3) The shear stress distributions along the flow section are different. The viscous shear stress is dominant near the wall (called the viscous sublayer).



(2) Viscous shear stress τ_v

Starting from the concept of time-averaged turbulence, there exists viscous shear stress between liquid layers:

$$\tau_v = \mu \frac{d\bar{u}}{dy}$$

$\frac{d\bar{u}}{dy}$ — Time-average velocity gradient

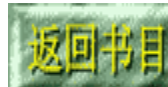
(3) Turbulent shear stress τ_t

The pulsation of the liquid particles leads to mass exchange, resulting in momentum exchange and mixing of particles, which further produces the turbulent shear stress at the interfaces of liquid layers:

$$\tau_t = -\rho \overline{u'_x u'_y}$$

derivation

Note: Turbulent shear stress is caused by the inertia of micro-cluster of fluids. It is only related to **fluid density** and **pulsation intensity**, but not directly related to **fluid viscosity**.



Derivation of $\overline{\tau_t} = -\rho \overline{u'_x u'_y}$



From the momentum equation, the momentum increment equals to the impulse generated by the turbulent shear force ΔT . That is:

$$\Delta T \cdot \Delta t = \Delta m u'_x = \rho \Delta A_x u'_y \Delta t \cdot u'_x$$

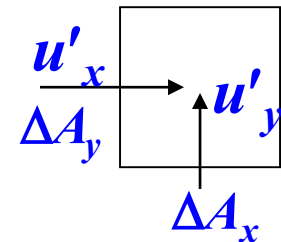
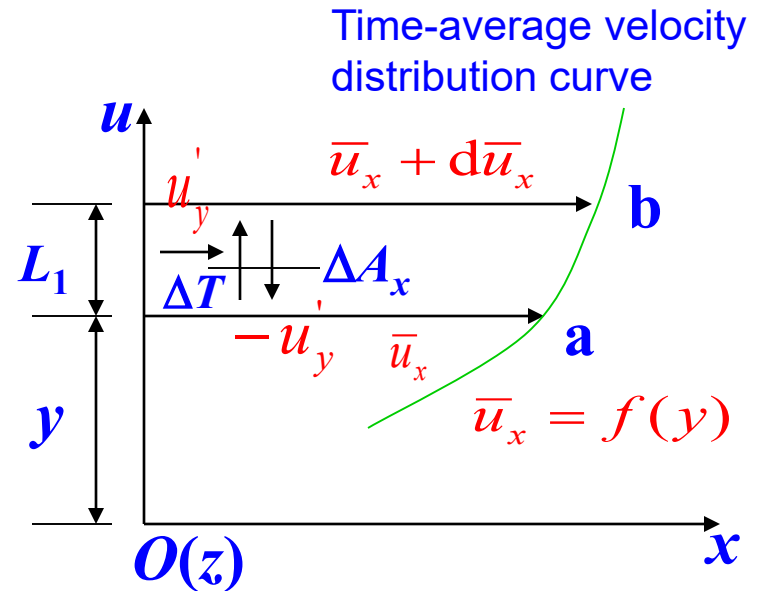
$$\tau_t = \frac{\Delta T}{\Delta A_x} = \rho u'_x u'_y$$

$$\overline{\tau_t} = \frac{1}{T} \int_0^T \rho u'_x u'_y dt = \rho \overline{u'_x u'_y}$$

Mass conservation gives: $u'_x \Delta A_y + u'_y \Delta A_x = 0$

$$\therefore u'_x = -\frac{\Delta A_y}{\Delta A_x} u'_y, \text{ } u'_x \text{ and } u'_y \text{ are in opposite symbols/direction}$$

$\overline{\tau_t} = -\rho \overline{u'_x u'_y}$ — expression of shear stress for two-dimensional turbulent flow



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4. Prandtl mixing length semi-empirical turbulence theory

a. Prandtl's hypothesis:



(1) The free path of molecular says: $\propto L_1 \frac{d\bar{u}_x}{dy}$ $u'_y \propto L_2 \frac{d\bar{u}_x}{dy}$

(2) Normalization gives: $\tau_t = -\rho \overline{u'_x u'_y} = \rho L^2 \left(\frac{d\bar{u}_x}{dy} \right)^2$

L is known as the mixing length, but without direct physical meaning. At or near the solid boundary of a turbulent flow, **Prandtl assumed that the mixing length is proportional to the radial distance from the particle to the wall**, i.e.,

$$L = \kappa y$$

where κ is a **dimensionless constant** determined by experiments.

E.g., $\kappa = 0.4$ for tube laminar flow. y is the distance to wall.

b. Expression for turbulent shear stress $\tau = \mu \frac{d\bar{u}_x}{dy} + \rho L^2 \left(\frac{d\bar{u}_x}{dy} \right)^2 = (\mu + \eta) \frac{d\bar{u}_x}{dy}$

$\eta = \rho L^2 \left(\frac{d\bar{u}_x}{dy} \right)$ is the **eddy viscosity** (涡粘性) .

- A property indicating the momentum transfer between turbulent particles
- Not dependent on fluid viscosity, but on fluid flow regime and fluid density

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Logarithmic distribution

Prandtl's hypothesis:

- (1) The shear stress τ near the wall is a constant and its value is equal to the shear stress at the solid wall τ_0 , i.e., $\tau = \tau_0$;
- (2) The mixing length l varies linearly with the distance y from the solid wall, i.e., $l = \kappa y$

Therefore
$$\tau = \tau_0 = \rho l^2 \left(\frac{d\bar{u}_x}{dy} \right)^2 = \rho \kappa^2 y^2 \left(\frac{d\bar{u}_x}{dy} \right)^2$$

$$u_* = \sqrt{\frac{\tau_0}{\rho}} = \kappa y \frac{d\bar{u}_x}{dy}$$

$$\frac{d\bar{u}_x}{u_*} = \frac{1}{\kappa} \frac{dy}{y}$$

$$\frac{\bar{u}_x}{u_*} = \frac{1}{\kappa} \ln y + C \quad (y > \delta_1)$$

Note: In the turbulent core region ($y > \delta_1$), the flow velocity is logarithmically distributed.

1. Why is the concept of **time averaging** introduced in studying turbulence?

In the turbulent flow regime, how to define steady and unsteady flows?

- With time averaging, the turbulent flow motion is simplified to the time-averaged flow without fluctuations, such that the time-averaged flow and the turbulent components can be studied separately.
- A flow in the turbulent regime is steady flow as long as the time-averaged quantities do not change with time.

2. What are the definitions and correlations of instantaneous velocity, turbulent velocity, time-averaged velocity and cross-sectional mean velocity?

- Instantaneous velocity u : the actual flow velocity of a fluid through a spatial location, fluctuating with time in the turbulent flow regime;
- Time-average velocity : the time average of the instantaneous velocity at a spatial location within time T , i.e., $\bar{u} = \frac{1}{T} \int_0^T u dt$
- Fluctuating velocity : the difference between instantaneous velocity and time-averaged velocity, i.e., $u = \bar{u} + u'$
- Cross-sectional mean velocity v is the cross-sectional average of velocities (the time-averaged velocity is considered for turbulent flow) at all locations of the flow section, i.e., $v = \frac{1}{A} \int_A \bar{u} dA$

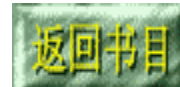


3. What are the **two forms of shear stress for turbulence**? What factors are they related to? Which regions do they mainly act in? 18

- Viscous shear stress: mainly related to fluid viscosity and velocity gradient between liquid layers. It mainly acts near the wall.
- Turbulent shear stress: mainly related to the pulsation intensity of fluid and the density of fluid, mainly acting in the turbulent core region where pulsation intensity is large.

4. Why the **viscous sublayer** exists in turbulent flow? What factors is the thickness related to? What is the significance for the analysis of turbulence?

- Near the wall, liquid particles are constrained by the solid wall and cannot undergo lateral motions. There is no mixing and the velocity gradient du/dy is very large. The viscous shear stress $\tau = \mu du/dy$ still plays the dominant role.
- The thickness of viscous sublayer is related to Re and the ability of particle mixing. With the increase of Re , the thickness decreases. The viscous sublayer is generally thin but has a significant influence on energy loss.



5. What are the differences between the division of **fluid layers** and the differentiation of **flow regimes** on a cross section of turbulent flow?

Viscous sublayer VS turbulence core: Viscosity, velocity distribution and velocity gradient on a cross section

Laminar flow VS turbulent flow: cross-sectional averaged velocity and Reynolds Number

6. What are the scope of application for the concepts of “structure of turbulence; time average of turbulent motion; Reynolds shear stress in turbulent flow; Prandtl’s semi-empirical mixing length theory and the distribution of turbulent flow velocity”?

(1) Pipe flow;

(2) Open-channel flow;

(3) Both pipe flow and open channel flow.



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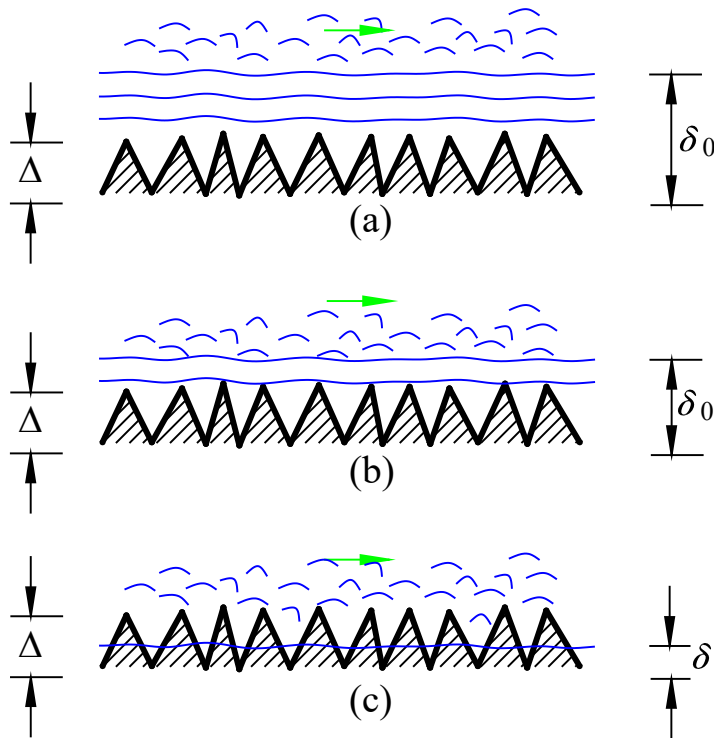
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(3) Reasons for the variation of turbulent resistance



Smooth turbulent flow

Transition turbulent flow

Rough turbulent flow

Extended reading: <https://dept.aem.umn.edu/~./faculty/joseph/PL-correlations/docs-In/R1-JFM-Submission-Virtual-Nikuradse.pdf>



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2. Turbulent velocity distribution in circular pipes

(1) Smooth region of turbulent flow

- Viscous sublayer

$$u = \frac{\tau_0}{\mu} y = \frac{\tau_0}{\rho} \cdot \frac{y}{\nu} = v_*^2 \cdot \frac{y}{\nu} \quad (y \leq \delta_0)$$

- Core region of turbulent flow

$$\frac{u}{v_*} = \frac{1}{\kappa} \ln y + C$$

With boundary conditions $y = \delta_0, u = u_b$ and $\delta_0 = \frac{v_b}{v_*} \cdot \frac{\nu}{v_*}$, we have:

$$\frac{u}{v_*} = \frac{1}{\kappa} \ln \frac{y v_*}{\nu} + C_1$$

From Nikuradse's experiment, $\kappa = 0.4, C_1 = 5.5$
it can be determined that

$$\frac{u}{v_*} = 5.75 \lg \frac{y v_*}{\nu} + 5.5 \quad (y > \delta_0)$$

——Semi-empirical formula for the velocity distribution
in the smooth turbulent region.



(2) Rough region of turbulent flow

$$\frac{u}{v_*} = 5.75 \lg \frac{y}{\Delta} + 8.48 \quad (y > 0)$$

Conclusion: The fluid velocities on the cross section of turbulent flow in circular pipe conforms to logarithmic distribution. (Those of laminar flows follow the parabolic surface)

(3) Exponential formula of the turbulent velocity distribution in circular pipe

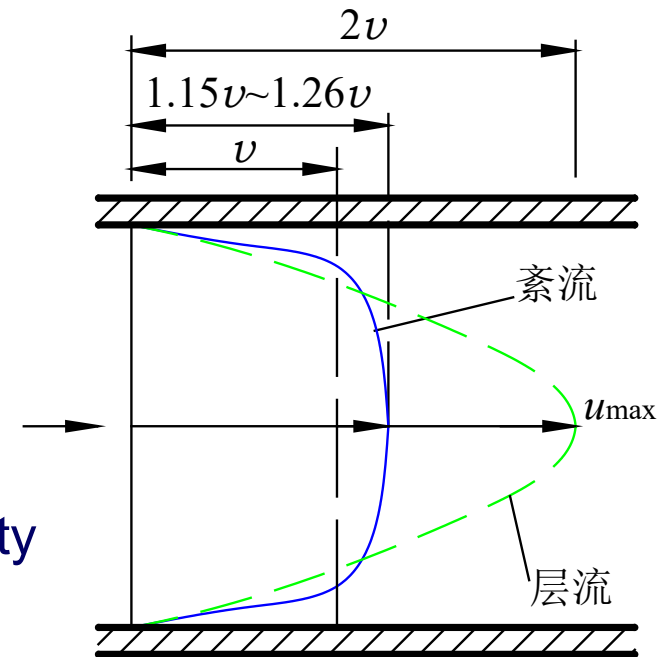
According to experimental data, Prandtl - Karman derived the **exponential formula** of turbulent velocity

distribution in circular pipes: $\frac{\bar{u}_x}{\bar{u}_{\max}} = \left(\frac{y}{r_0}\right)^n$ n is the index decreasing with the increase of Re .

For smooth pipes:

1) When $Re = 1.1 \times 10^5$, $n = \frac{1}{7}$, $\frac{\bar{u}_x}{\bar{u}_{\max}} = \left(\frac{y}{r_0}\right)^{1/7}$, it is called one seventh law of turbulent velocity distribution.

2) Increasing Re , n decreases. When $Re \geq 2 \times 10^6$, $n = 1/10$; when $Re \geq 4 \times 10^3$ & $Re < 2 \times 10^6$, n calculated by a empirical formula.



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3. Semi-empirical formula for turbulent flow in circular pipes

Nikuradse semi-empirical formula

a. Smooth turbulent flow region $\frac{1}{\sqrt{\lambda}} = -2\lg\left(\frac{2.51}{Re\sqrt{\lambda}}\right)$

b. Rough turbulent flow region $\frac{1}{\sqrt{\lambda}} = -2\lg\left(\frac{\Delta}{3.7d}\right)$

4. Industrial pipelines: Barr formula, Moody diagram

- Industrial pipeline: Practical pipelines without manual roughening.
- Equivalent roughness: The wall roughness of industrial pipelines obtained by experiments through comparing Nikuradse's diagram.

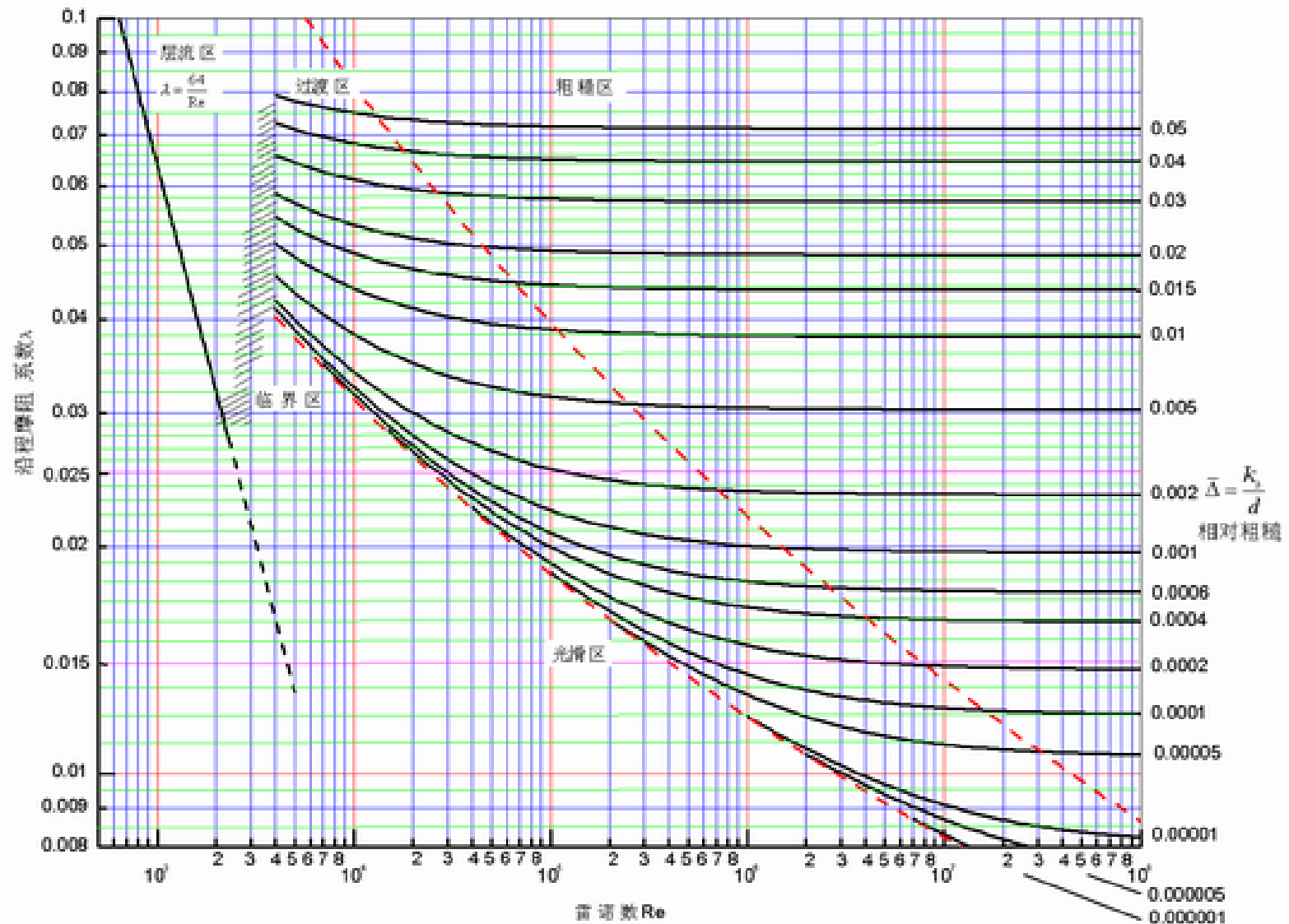
Colebrook-White equation $\frac{1}{\sqrt{\lambda}} = -2\lg\left(\frac{2.51}{Re\sqrt{\lambda}} + \frac{\Delta}{3.7d}\right)$ ——— Moody diagram

Barr formula $\frac{1}{\sqrt{\lambda}} = -2\lg\left(\frac{5.1286}{Re^{0.89}} + \frac{\Delta}{3.7d}\right)$ An explicit formula

Scope of applications: applicable to smooth and rough turbulence regimes, as well as the transition regime of circular-pipe turbulence



Moody Diagram



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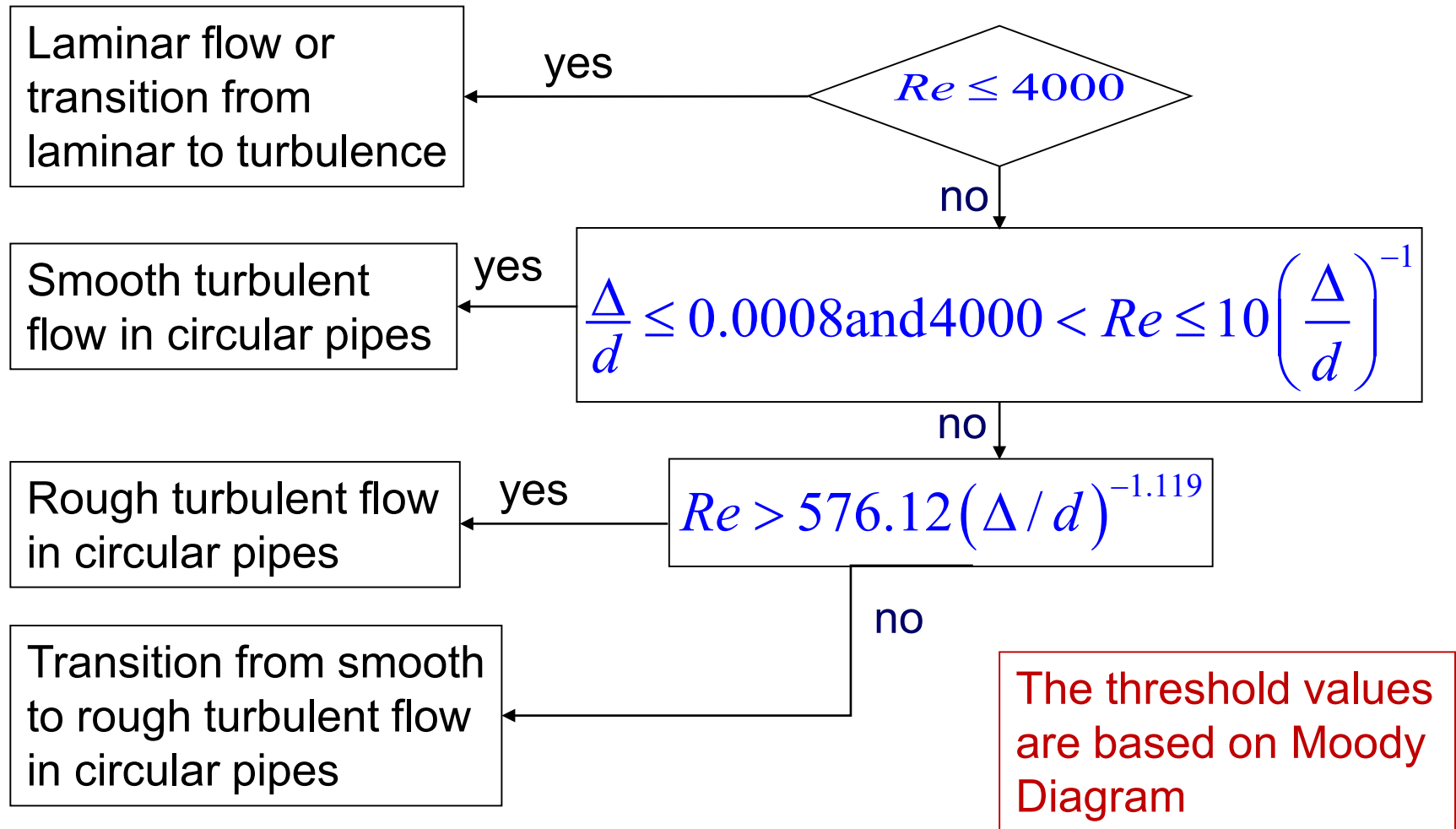
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5. Identify flow resistance zones of turbulent flow



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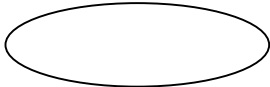
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1. Pressurized flow in non-circular pipes

$$h_f = \lambda \frac{l}{d_e} \frac{v^2}{2g}$$

$d_e = 4R$, is the equivalent diameter of non-circular pipes,
applicable to:

Rectangle: b  a $a/b < 8$

Ellipse:  Long axis / short axis < 3

Example1

Example2

Example3



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2. Open-channel flow

Chezy formula (谢才公式): $v = C\sqrt{RJ}$ $\longrightarrow$ $h_f = \frac{l}{R} \frac{v^2}{C^2} = \frac{8g}{C^2} \frac{l}{4R} \frac{v^2}{2g}$

C : Chezy coefficient with the dimension of, $\text{m}^{0.5}/\text{s}$

$C = \sqrt{\frac{8g}{\lambda}}$ $\longleftarrow$ $\lambda = \frac{8g}{C^2}$

- Manning formula (曼宁公式):

$$C = \frac{1}{n} R^{1/6}, \quad n \text{ — roughness factor}$$

- Empirical formula, dimensional **non-homogeneity**
- Cast iron pipe, steel pipe $n=0.011 \sim 0.013$; Concrete pipe $n=0.011 \sim 0.014$

Example 1: Pipe length $l = 500\text{m}$, diameter $d = 0.2\text{m}$, height of pipe wall rough protrusion $\Delta = 0.1\text{mm}$. With a flow rate $q_v = 10 \times 10^{-3} \text{ m}^3/\text{s}$ and the water temperature $T = 10^\circ\text{C}$, calculate the frictional head loss?

Solution:
$$v = \frac{q_v}{\frac{1}{4}\pi d^2} = \frac{10 \times 10^{-3}}{\frac{1}{4}\pi (0.2)^2} = 0.3183 \text{ m/s}$$

$\because t = 10^\circ\text{C} \quad \therefore \nu = 1.310 \times 10^{-6} \text{ m}^2/\text{s}$

$$Re = \frac{vd}{\nu} = \frac{0.3183 \times 0.2}{1.31 \times 10^{-6}} = 48595 > 2300 \quad \text{So turbulent flow in the pipe.}$$

Barr formula:
$$\frac{1}{\sqrt{\lambda}} = -2 \lg \left(\frac{\Delta}{3.7d} + \frac{5.1286}{Re^{0.89}} \right)$$

Substituting $\frac{\Delta}{d} = 0.0005$, $Re = 48595$ gives $\lambda = 0.0227$

Therefore,
$$h_f = \lambda \frac{l}{d} \frac{v^2}{2g} = 0.0227 \times \frac{500}{0.2} \times \frac{0.3183^2}{2 \times 9.8} = 0.293 \text{ m}$$

$$A = \frac{\pi}{4} d^2 = 0.785 \times 0.4^2 = 0.126 \text{m}^2$$
$$R = \frac{A}{\chi} = \frac{0.785d^2}{\pi d} = \frac{d}{4} = 0.1\text{m}$$
$$C = \frac{1}{n} R^{\frac{1}{6}} \quad C = \frac{1}{0.011} 0.1^{\frac{1}{6}} = 61.94 \text{m}^{1/2}/\text{s}$$
$$q_v = vA = AC\sqrt{RJ}, \quad J = \frac{h_f}{l}$$

$$q_v = 0.126 \times 61.94 \sqrt{0.1 \times \frac{0.4}{100}} = 0.156 \text{ m}^3/\text{s}$$

Example 3: A circular-section tunnel with concrete lining: diameter $d=2000\text{mm}$, pipe length $l=100\text{m}$, flow rate $q_v=31.4\text{m}^3/\text{s}$, roughness $n=0.014$. Solve for the frictional head loss h_f .

Solution: The flow in a tunnel is generally rough turbulent flow. So the Chezy formula can be applied.

So

$$q_v = CA\sqrt{RJ} = CA\sqrt{R\frac{h_f}{l}}$$

$$h_f = \frac{q_v^2 l}{C^2 A^2 R}$$

where $A = \frac{\pi}{4}d^2 = 0.785 \times 2^2 = 3.14\text{m}^2$

$$R = \frac{d}{4} = 0.5\text{m}$$

$$C = \frac{1}{n}R^{\frac{1}{6}} = \frac{1}{0.014} \times 0.5^{\frac{1}{6}} = 63.6\text{m}^{\frac{1}{2}}/\text{s}$$

Substituting the above values into the formula of h_f gives,

$$h_f = \frac{31.4^2 \times 100}{63.6^2 \times 3.14^2 \times 0.5} = 4.94\text{m}$$

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